

EXPERIMENT LIFECYCLE

1. **Hypothesis and decision:** what will we do differently if it wins?
 2. **Metrics:** one primary (the decision metric), a few secondary, **guardrails** that must not worsen (latency, refunds, unsubscribes).
 3. **Unit of randomisation:** usually the user; analyse at the same unit. Exclusions defined up front.
 4. **Sample size, duration** from baseline, minimum detectable effect (MDE), α and power.
 5. **Validate setup:** A/A test, sample ratio mismatch (SRM), balance on pre-period metrics.
 6. **Run** the planned duration, whole weeks, no peeking decisions.
 7. **Analyse** with the pre-registered test; check guardrails and segments (as hypotheses).
 8. **Decide, document, monitor** after launch (holdout, long-term effect).
- Write the decision rule beforehand: ship if lift $\geq$ MDE and guardrails flat; iterate if direction good but inconclusive; stop otherwise.

METRIC TYPES

Type	Example	Note
Primary / north star	conversion, revenue per user	one, sensitive, tied to decision
Driver / diagnostic	click-through, add-to-cart	explain the primary
Guardrail	latency, crash rate, refund rate, retention	must not degrade
Ratio	CTR = clicks/impressions	unit mismatch: use per-user or delta method

- Good metrics: measurable, sensitive, hard to game, aligned with long-term value. Optimising one alone gets gamed (clicks up, satisfaction down).

- Per-user metrics with heavy tails (revenue): winsorise, log-transform, or use a bootstrap.

SAMPLE SIZE (PER GROUP)

- Two proportions: $n = (z\alpha/2 \cdot \sqrt{p(1-p)} + z\beta \cdot \sqrt{p_1(1-p_1) + p_2(1-p_2)})^2 / \delta^2$, with p the pooled rate and $q = 1 - p$. Rule of thumb: $n \sim 16 \cdot p(1-p) / \delta^2$ ($\alpha = 0.05$, power 80%).
- Means: $n \sim 16\sigma^2 / \delta^2$ where δ is the absolute difference.
- z values: $\alpha = 0.05$ two-sided 1.96; power 80% 0.84; power 90% 1.28.

Baseline	+5% rel	+10% rel	+20% rel
2%	315k	81k	21k
5%	122k	31k	8.2k
10%	58k	15k	3.8k
20%	26k	6.5k	1.7k
50%	6.3k	1.6k	390

- Halving the MDE needs 4x the users. Duration = $(2n) / (\text{daily eligible users})$, rounded up to full weeks.
- Variance too high? Use CUPED, a more sensitive metric, a larger MDE, or longer duration.

RANDOMISATION AND EXPOSURE

- Assign by hashing the user id with an experiment salt: stable across visits, independent across experiments. Never assign by session if the metric is per user.
- **Ramp** 1% $\rightarrow$ 10% $\rightarrow$ 50% to catch bugs; the analysis uses the full-allocation period only.
- Log **exposure** (when a user actually sees the change) and analyse the triggered population to cut noise; keep the intent-to-treat view too.
- **Layers / mutually exclusive** experiments avoid interaction between concurrent tests. Keep a **global holdout** for long-term effects.
- Balance check: pre-experiment metrics and device mix should be equal across arms.

- Hypotheses: H_0 no difference; H_1 two-sided unless a one-sided direction is justified in advance.

ANALYSIS STEPS

1. Check SRM first: chi-square of observed vs expected split. 257.5k/242.5k out of 500k, expected 50/50: chi-square = 450, p tiny: a bug, do not trust the results.
2. Compute lift with a **confidence interval**, not just p.
3. Two proportions: $z = (p_2 - p_1) / \sqrt{p(1-p)(1/n_1 + 1/n_2)}$, p pooled. Example: 10% vs 11%, 14,400 each: $z = 2.77$, $p = 0.006$.
4. Means: Welch t-test; heavy tails: bootstrap CI.
5. Guardrails and secondary metrics: expect noise; correct for many metrics.
6. Segments (platform, new vs returning) only as hypotheses, or pre-planned.
7. Practical significance: is the CI above the minimum worth shipping?

```
import statsmodels.stats.proportion as sp
z, p = sp.proportions_ztest([c2, c1], [n2, n1])
es = sp.proportion_effectsize(0.11, 0.10)
from statsmodels.stats import power
power.NormalIndPower().solve_power(es, power=0.8, alpha=0.05)
```

VARIANCE REDUCTION AND RATIO METRICS

- **CUPED:** $Y_{\text{adj}} = Y - \theta(X - \text{mean } X)$, $\theta = \text{cov}(X, Y) / \text{var}(X)$, X = pre-experiment metric. Variance shrinks by $(1 - r^2)$: $r = 0.7$ halves it.
- **Stratification / blocking** by platform or country; **paired designs** where possible.
- **Ratio metrics** (CTR, revenue per session): analyse per user or use the delta method; the unit of analysis must match the unit of randomisation.
- Winsorise at the 99th percentile for revenue; report both.

PITFALLS AND FIXES

Pitfall	Fix
Peeking and stopping at first $p < 0.05$	fixed horizon or sequential testing
Multiple metrics / segments	one primary, correct (BH), hypotheses
SRM (split off)	find the bug (bots, redirects, logging) and rerun
Novelty / primacy effect	run 2+ weeks, look at later cohorts
Underpowered test	compute n ; report CI, not "no effect"
Interference (marketplaces, social)	cluster, geo or switchback designs
Wrong unit (session vs user)	analyse at the randomisation unit
Seasonality, launches	full weeks, avoid holidays, run concurrently
Survivorship (analyse only active users)	intent-to-treat analysis
Simpson's paradox	stratify, check allocation by segment

- Weekday effects: run whole weeks. Carry-over: do not reuse exposed users.
- A/A tests should reject ~5% of the time at $\alpha = 0.05$.

ADVANCED DESIGNS

- **Sequential testing** (alpha spending, mSPRT): valid early looks. **Bayesian A/B**: $P(B > A)$ and expected loss; choose a prior; still need a stopping plan.
- **Multi-armed bandits** (epsilon-greedy, Thompson sampling): shift traffic to winners, less regret, weaker inference. Good for short-lived choices (headlines).
- **Cluster randomisation**: randomise groups (cities, classrooms); effective sample size is smaller (design effect = $1 + (m - 1) \cdot ICC$).
- **Switchback**: alternate treatment by time block in one market (pricing, dispatch).
- **Quasi-experiments** when you cannot randomise: **difference-in-differences** (treated change – control change, needs parallel trends), **synthetic control**, regression discontinuity, matching / propensity scores, instrumental variables.

- **Interleaving** for ranking and search; **holdouts** for long-term effects; **multivariate** tests need much more traffic.
- **Uplift modelling**: who changes behaviour because of treatment, not who is likely to convert anyway.
- **Network effects**: randomise by graph cluster; expect bias if you do not.

DECISION MATRIX

Result	Action
CI above MDE, guardrails flat	ship (maybe ramp 10 → 50 → 100%)
CI includes 0, adequate power	no detectable effect: stop or iterate
CI includes 0, underpowered	extend or redesign, do not call it flat
Primary up, guardrail down	trade-off: quantify, product call
Segment-only win	follow-up test, do not ship on it
SRM or tracking bug	invalid: fix and rerun

- Always say: effect size, interval, cost of being wrong, next step.

CASE-ANSWER SKELETON

1. Clarify the goal and the decision.
 2. Primary metric, guardrails, unit of randomisation, exclusions.
 3. Sample size, duration, ramp plan.
 4. Risks: novelty, interference, seasonality, instrumentation, SRM.
 5. Analysis plan: test, CI, segments, corrections.
 6. Decision rule for each outcome; follow-up and monitoring.
- **Cannot A/B test?** Staged rollout with holdout, difference-in-differences, synthetic control, or before-and-after with caveats.
 - **Two-sided marketplace pricing**: randomise by market or time; track both sides (rider wait, driver earnings, cancellations).
 - **Only 5,000 users a week, heavy-tailed revenue**: CUPED, winsorise, a proxy metric, larger MDE, longer run.
 - **+0.3% conversion but AOV down**: judge revenue per user with its CI.

PRE-LAUNCH AND REVIEW CHECKLIST

- Hypothesis, decision rule and primary metric written down. Sample size, duration, MDE agreed.

- Instrumentation tested: events fire once, in both arms, with the experiment id.
- Eligibility and exclusions (bots, employees, repeat users) defined; exposure logged.
- Guardrails and alerts (auto-stop on crashes or revenue collapse).
- After: SRM, novelty check (effect by week), CI, segments flagged as exploratory, write-up with the decision and learning.
- Tell stakeholders: "We estimate +X% (95% CI a to b). Ship because ... Risk ...".

WHERE TESTS GO WRONG IN PRACTICE

- Tracking changes in one arm only; caching, redirects, or app versions that never see the variant.
- Treatment that changes who is counted in the metric (more users reach checkout, conversion falls): use a denominator that is unaffected.
- Mixed-up control (existing users) vs treatment (new users): not a randomised comparison.
- Effects that decay (promotions), or build slowly (habit features): use cohorts and a holdout.
- Testing many small variants at once: low power and multiple-testing risk.

QUICK NUMBERS

- 95% CI half-width for a proportion $\sim 2 \cdot \sqrt{p(1-p)/n}$: $n = 10,000$, $p = 10\%$: $\pm 0.6pp$.
- Detect 1pp on a 10% base needs ~14,400 per group; 0.5pp needs ~57,800.
- 20 metrics at $\alpha = 0.05$: 64% chance of at least one false positive.
- Dates: avoid holidays, launches; at least one full business cycle.

SAY IT IN 20 SECONDS

- "Randomise at the user, size for the effect that matters, run full weeks, no peeking."
- "A significant lift with a falling guardrail is not a win."
- "SRM means stop and debug; do not interpret."
- "No significance with low power is not 'no effect'."